**Suggested problems:** 1-15 odd

**Problem 1:** Find $dz/dt$.

(a) $z = x^2y - y^2$, $x = \sin(t)$, $y = e^t$

(b) $z = xy^2$, $x = e^{-t}$, $y = \sin(t)$

(c) $z = x\sin(y) + y\sin(x)$, $x = t^2$, $y = \ln(t)$

**Problem 2:** Let $z = f(x, y)$ where $x = g(t)$ and $y = h(t)$. Given the information below, find $dz/dt|_{t=1}$.

| | |
|---|---|
| $f(3, 10) = 7$ | $f(4, 11) = -20$ |
| $f_x(3, 10) = 100$ | $f_y(3, 10) = 0.1$ |
| $f_x(4, 11) = 200$ | $f_y(4, 11) = 0.2$ |
| $f(3, 4) = -10$ | $f(10, 11) = -1$ |
| $g(1) = 3$ | $h(1) = 10$ |
| $g'(1) = 4$ | $h'(1) = 11$ |

**Problem 3:** A steel bar with square cross sections 5 cm by 5 cm and length 3 meters is being heated. For each dimension, the bar expands $13 \times 10^{-6}$ meters for each 1°C rise in temperature. What is the rate of change in the volume of the steel bar?

**Problem 4:** Find $\partial z/\partial u$ and $\partial z/\partial v$.

(a) $z = 2xy$, $x = u^2 + v^2$, $y = u/v$

(b) $z = x^2 e^y$, $x = 4u$, $y = 3u^2 - 2v$